Tutorial 2 - Conda Environment, Indentation & Type Casting, Booleans, Conditional Logic, Evaluation Flow, Operator Precedence, Loops

Section A: Multiple Choice Questions (MCQ)

1. Which command creates a new conda environment named 'mlenv' with Python 3.12?

A. conda create mlenv python=3.12
B. conda create -n mlenv python=3.12
C. conda install mlenv python=3.12
D. python create mlenv=3.12

Answer: B

2. Which command activates a conda environment?

A. conda start env
B. conda run env
C. conda activate env_name
D. activate env_name

Answer: C

3. What happens if Python indentation is incorrect?

A. Code runs slower
B. SyntaxError
C. LogicalError
D. Nothing

Answer: B

4. What is the output of int('5') + 2?

A. '52'
B. 7
C. Error
D. '7'



Answer:

5. Which of the following is a Boolean value?

A. 'True'
B. TRUE
C. True
D. 1



Answer:

6. What is the result of True and False?

A. True
B. False
C. None
D. Error



Answer:

7. Which collection type is mutable?

A. Tuple
B. List
C. String
D. None



Answer:

8. Which collection does NOT allow duplicate values?

A. List
B. Tuple
C. Set
D. Dictionary



Answer:

9. How do you access the first element of a list 'a'?

A. a(0)
B. a[1]
C. a[0]
D. a{0}



Answer:

10. Which data structure stores key-value pairs?

A. List
B. Tuple
C. Set
D. Dictionary



Answer:

11. What is the output of len([1,2,3])?

A. 2
B. 3
C. 4
D. Error



Answer:

12. Which of the following is immutable?

A. List
B. Dictionary
C. Tuple
D. Set

C

Answer:

13. What is the result of [1,2,3][1:3]?

A. [1,2]
B. [2,3]
C. [1,2,3]
D. [2]

B

Answer:

14. What does set([1,1,2,2,3]) return?

A. [1,1,2,2,3]
B. {1,2,3}
C. (1,2,3)
D. Error

B

Answer:

15. Which keyword is used for conditional logic?

A. loop
B. if
C. switch
D. case

B

Answer:

16. What is evaluated first? 3 + 2 * 5

A. 3+2
B. 2*5
C. 3*2
D. None



Answer:

17. Which loop runs at least once?

A. for loop
B. while loop
C. do-while loop
D. None



Answer:

18. What is the output of: for i in range(3): print(i)?

A. 1 2 3
B. 0 1 2
C. 0 1 2 3
D. 1 2



Answer:

Section B: Fill in the Blank (Programming)

1. conda _____ -n testenv python=3.12

Answer: create

2. conda _____ testenv

Answer: activate

3. x = int('10') → x = _____

Answer: 10

4. y = float('3.5') → y = _____

Answer: 3.5

5. True and True → _____

Answer: True

6. False or True → _____

Answer: True

7. if x > 5: print(x) → keyword is _____

Answer: if x>5

8. 3 + 2 * 2 → _____

Answer: 7

9. for i in _____(5):

Answer: range

10. while x < 5: x _____ 1

Answer: +

Section C: Coding Troubleshooting & Logic

1. The following Python code is intended to print the value of x if it is greater than 5, but it produces an error.

```
if x > 5
   print(x)
```

Analyze the code and:

a. Identify the cause of the error
b. Explain why the error occurs
c. Provide the corrected version of the code

a) There is no colon after x>5
b) This is a syntax error since the if loop is not declared properly.
c) if x>5:
    print(x)

2. The following Python code produces an error:

```
x = '5'
print(x + 2)
```

Analyze the code and:

a. Identify the cause of the error
b. Explain why the error occurs
c. Provide the corrected version of the code

a) It is because of x = '5'
b) It is a syntax error, this is because x in this case is a string. Therefore, when the user is printing (x+2), it produces an error because you cannot concatenate a string and number.
c) int x = '5'
    print (x+2)

3. Why does this loop run forever?

```
while x < 5:
   print(x)
```

Answer: This loop runs forever if x is < 5, there is no operators in the loop that will change the value of x, causing it to run forever

4. What is wrong?

```
for i in range(5)
   print(i)
```

Answer: There is no colon at the end of range(5)

5. Predict output:

```
x = 2
if x > 1 and x < 3:
   print('Yes')
```

Answer: Yes